MATH-101 Calculus and Analytical Geometry – 3+0 CHs

By

Prof Dr Safia Akram

TEXTBOOK:

Thomas Finney 13th Edition Calculus and Analytical Geometry

REFERENCE BOOK:

a) Howard Anton, Irl C. Bivens, Stephen Davis, Calculus 10th Edition

b) Swokowski, Onlinick & Pence: Calculus (6th Edition)

c) Robert T. Smith & Roland B. Minton: Calculus (3rd Edition)

Course Contents:

S/No	Topics
1.	Review of vectors, scalars and vector products. Three-dimensional coordinate system and equation of straight line and plane
2.	Limits & continuity, techniques of finding limits
3.	Techniques of differentiation, Tangent lines and rates of change
4.	Extrema of functions, Roll's and Mean value theorems, Concavity
5	Riemann sum, definite integrals and properties of integrals
6.	Solids of revolution, volume of solids of revolution by Cylindrical shell & Cross section methods
7.	Arc length, surface of revolution, Center of mass
8.	Indeterminate forms and L Hospital rule, trigonometric integrals.
9.	Improper Integrals
10.	Convergence and divergence of sequences and series, positive term series, integral test
11.	Basic comparison test, limit comparison test, the ratio and root tests, alternating series, absolute and conditional convergence
12.	Power series, Maclaurin and Taylor series

Assigned PLO	Course Learning Outcomes CLOs	Level of Learning	PLO
1, 2	**CLO1** Interpret the basic concepts of analytical geometry	C2	1
	CLO2 Use the concepts of limits and continuity	C3	1
	CLO3 Apply techniques of differentiation and integration to engineering problems	C3	2

Program Learning Outcomes

PLO 1	**Engineering Knowledge**
	An ability to apply knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.
PLO 2	**Problem Analysis**
	An ability to identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.
PLO 3	**Design/Development of Solutions**
	An ability to design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.
PLO 4	**Investigation**
	An ability to investigate complex engineering problems in a methodical way including literature survey, design and conduct of experiments, analysis and interpretation of experimental data, and synthesis of information to derive valid conclusions.
PLO 5	**Modern Tool Usage**
	An ability to create, select and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modeling, to complex engineering activities, with an understanding of the limitations.

PLO 6	**The Engineer and Society**
	An ability to apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solution to complex engineering problems.
PLO 7	**Environment and Sustainability**
	An ability to understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate knowledge of and need for sustainable development.
PLO 8	**Professional Ethics**
	Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.

PLO 9	**Individual and Teamwork**
	An ability to work effectively, as an individual or in a team, on multifaceted and /or multidisciplinary settings.
PLO 10	**Communication**
	An ability to communicate effectively, orally as well as in writing, on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PLO 11	**Project Management**
	An ability to demonstrate management skills and apply engineering principles to one's own work, as a member and/or leader in a team, to manage projects in a multidisciplinary environment.
PLO 12	**Lifelong Learning**
	An ability to recognize importance of, and pursue lifelong learning in the broader context of innovation and technological developments

Course Assessment		
Exam:	1 Midterm 1 Final Examination	
Homework:	3 -- 4 Assignments	
Quizzes:	4 -- 5 Quizzes (Will be held on LMS)	
Grading:	Quizzes:	10%
	Assignments:	10%
	Midterm :	30%
	Final Exam:	50%

ATTENDANCE

75% Attendance is mandatory